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1. Project Objectives

1.1 Product Analysis: What are the products that generate the most sales revenue?

Product analysis is an important process for the management. By evaluating the overall performance of their products, it allows them to identify strengths, weaknesses and opportunities to stay competitive in the market. This analysis will help decide which products to continue, improve or discontinue. In addition, the management can have a greater control of their inventory and reduce storage costs by understanding which products that are bestsellers, slow movers and even dead stock. This process will provide the management with stronger competitive positioning to make smarter decisions.

1.2 Customer Analysis: Which customer segment (Consumer vs. Corporate) is most profitable?

By evaluating each customer segment, the online electronic retailer can tailor their strategies to the better-performing segment. If one segment really outperforms the other, then management may want to apply more attention and resources there. But supposing both are valuable to the company? They can also use a differentiated strategy for each. The management must know where profits come from so, they can concentrate on these matches and avoid wasting resources on non-performers.

1.3 Place Analysis: Which country/region has the most sales?

The management can identify top-performing markets to distribute their resources effectively and potentially plan for an expansion. Sales and profit geography enables the management to make smarter decisions on where to invest and grow the business. Based on profitability and growth prospects, they can choose where to extend more effort or target less. It might make sense to also establish regional warehouses for the areas with most sales in order to reduce shipping delays and costs. So, place analysis will provide information about where the company may be able to strategically invest in high-profit countries or regions.

2. Data Preparation

The given dataset is made available in xlsx format containing 20281 rows and 23 columns. It was generally complete with no missing values or records found in the dataset. However, there were duplicate records, inconsistent categorical values and logical errors. The table below are steps taken to clean up the data by its respective columns.

Column No.	Column Name	Description	Remarks / Cleaning Action Taken
1	Order ID	The unique number identifier generated when order is placed	It was identified that multiple customer orders were sharing the same Order ID. To resolve this, a new unique Order ID was assigned to all customers using a combination of the original Order ID, Order Date and Customer ID to maintain data integrity. (Example: 12-20190926-Yad10202)
2	Order Date	The date on which the order is placed	The Order Date must be before the Ship Date. To validate this, conditional formatting was applied to check whether Order Date < Ship Date. There were multiple dates that failed this validation. However, no data cleaning is required because this logical error may require further review from the party responsible before correcting this issue.
3	Ship Date	The date that the order is shipped from the seller or warehouse to the customer	The Ship Date must be after the Order Date. To validate this, conditional formatting was applied to check whether Order Date < Ship Date. There were multiple dates that failed this validation. However, no data cleaning is required because this logical error may require further review from the party responsible before correcting this issue.
4	Ship Mode	The shipping mode category	There are only 4 distinct values for this categorical column which are valid and consistent. No data cleaning is required.
5	Customer ID	The unique number identifier for each customer	There were customers of the same name that shared the same Customer ID. No data cleaning is required because Customer ID was likely generated without a unique identifier (e.g., email, phone number, or identification number),

			which was unavailable in this dataset.
6	Customer Name	The name of the customer	The Customer Name does not really matter if the Customer ID differs from each unique customer. Therefore, no data cleaning is required because it is common for multiple customers to share the same name.
7	Segment	Whether the customer is from consumer or Corporate	There are only 2 distinct values for this categorical column which are valid and consistent. No data cleaning is required.
8	City	The city where the order is to be delivered	The values for this column are valid and consistent. No data cleaning is required.
9	Latitude	Latitude of the location	The values for this column are valid and consistent. No data cleaning is required.
10	Longitude	Longitude of the location	The values for this column are valid and consistent. No data cleaning is required.
11	State	The State where the order is to be delivered	The values for this column are valid and consistent. No data cleaning is required.
12	Region	The Country region where the order is to be delivered	This column contains a mix of geographic data such as continent, region and sub-region. To improve consistency, a lookup table was used to map each country's ISO3 code to its corresponding region.
13	ISO3	ISO3 code of countries	The values for this column are valid and consistent, requiring no data cleaning. In fact, this column was used to map each country to its corresponding region.
14	Country	Country name	The values for this column are valid and consistent. No data cleaning is required.
15	Market	The market segment of the order	There should only be 4 distinct values for the market segment (APAC, EMEA, LATAM and NAR). Any additional values were mapped

			to these 4 categories using a lookup table.
16	Product ID	The unique number identifier for each product	The Product ID is unique, and both the Product ID and Product Name share the same last 4 digits to indicate their association. Therefore, no data cleaning is required.
17	Product Name	Product identified name	The Product Name can be non-unique since different products might share similar names. The values are valid and consistent, so no data cleaning is required.
18	Sub-Category	Product sub-category	There are only 2 distinct values for this categorical column which are valid and consistent. No data cleaning is required.
19	Category	Product Category	The values for this column are valid and consistent. However, there was a minor spelling error where "accessoires" was used instead of "accessories". This was corrected by using the find and replace function.
20	Quantity	Quantity of the product ordered	The values for this column are valid and consistent. No data cleaning is required.
21	Discount	Discount in decimal	The values for this column are valid and consistent. No data cleaning is required.
22	Sales	Net Sales of the order	The values for this column are valid and consistent. No data cleaning is required.
23	Profit	Net Profit of the order	The values for this column are valid and consistent. No data cleaning is required.

2.1 Data Type

In Tableau Data Source, all of the columns are automatically converted to their appropriate data types. Intentionally, columns such as Region and Sub Region will be treated as categorical labels because they are not actual geographic locations.

2.2 Mapping of Region, Sub Region and Market using a lookup table

A part of the lookup table will look as follows:

ISO3	Region	Sub Region	Market
MAR	Africa	Northern Africa	EMEA
AUS	Oceania	Australia and New Zealand	APAC
USA	Americas	Northern America	NAR
ARG	Americas	Latin America and the Caribbean	LATAM

This lookup table combines data from lukes' *ISO-3166 Countries with Regional Codes* repository on GitHub (luke, n.d.) and Revenera's *Countries by Region* table from the FlexNet Operations documentation (Revenera, n.d.). The mapping process was completed successfully, with consistent assignment of the region, sub region and market attributes based on the original ISO3 codes. An additional column sub region was created to go beyond broader region classification.

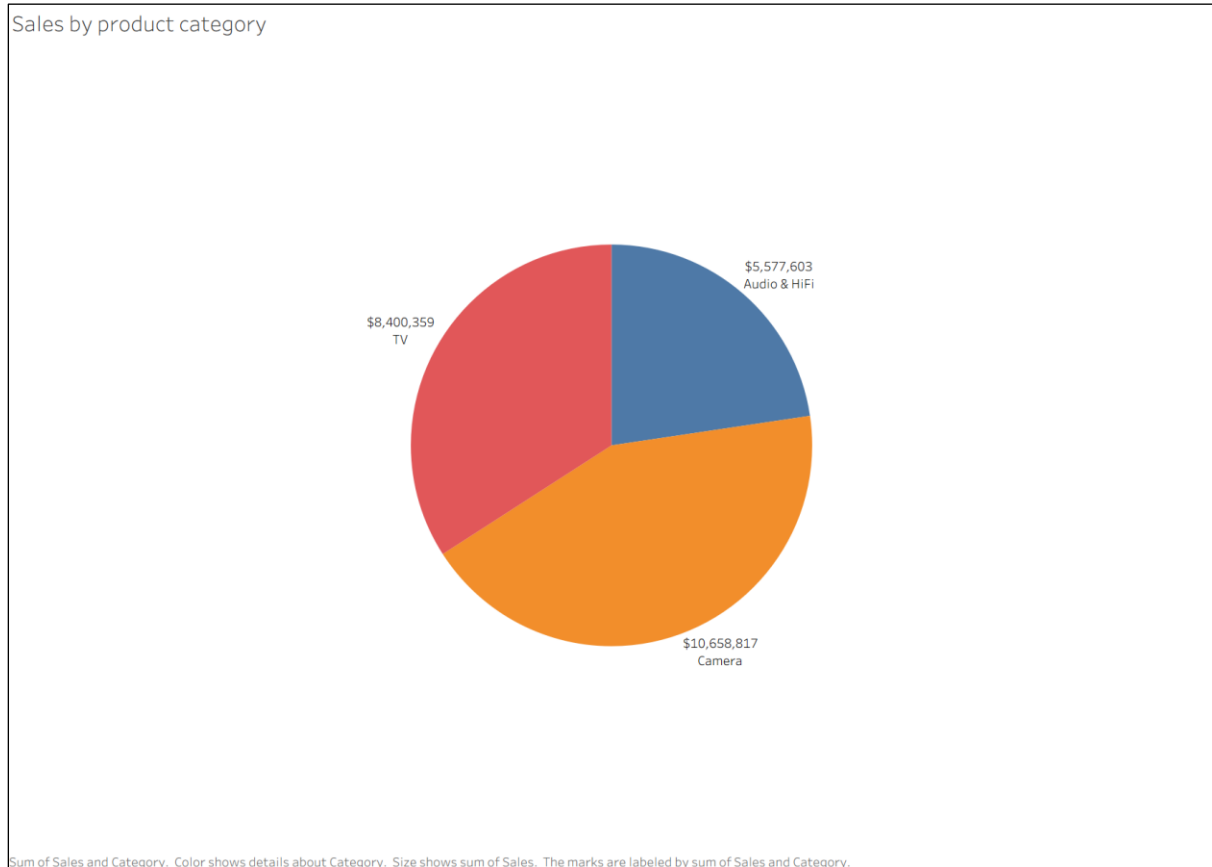
2.3 Creating a new calculated field (Profit Margin)

Profit margin is the ratio of profit to revenue, showing how much profit is earned for each dollar of revenue. Given that the dataset includes columns like sales and profit, introducing an additional metric can uncover insights for more effective price analysis rather than relying solely on raw sales numbers or profit figures.

3. Exploratory Data Analysis and Visualisation

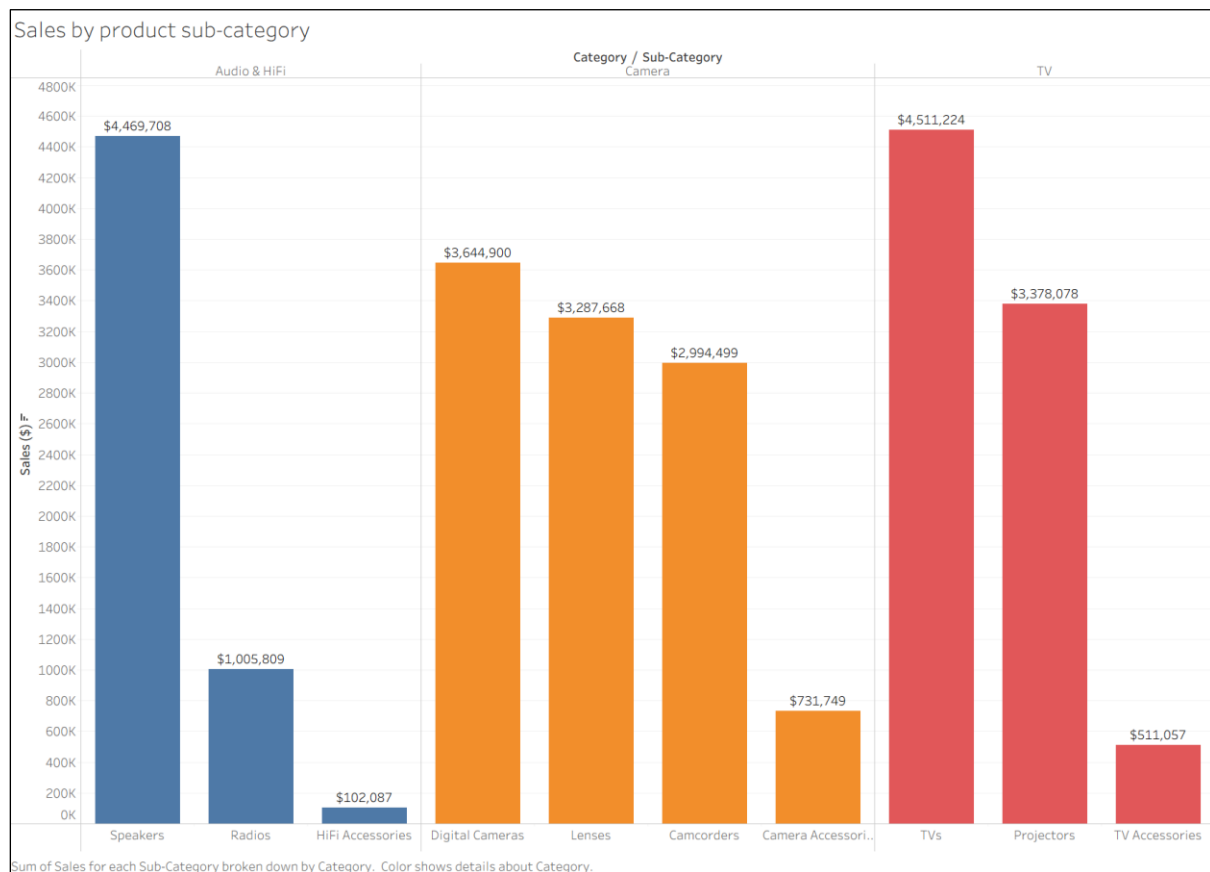
3.1 Sales by product category

This pie chart shows the sale distribution by product category. Camera is the top-selling category with the highest sales of \$10,658,817. TV is the second highest, with sales of \$8,400,359 and Audio & HiFi is the smallest slice with sales of \$5,577,603.



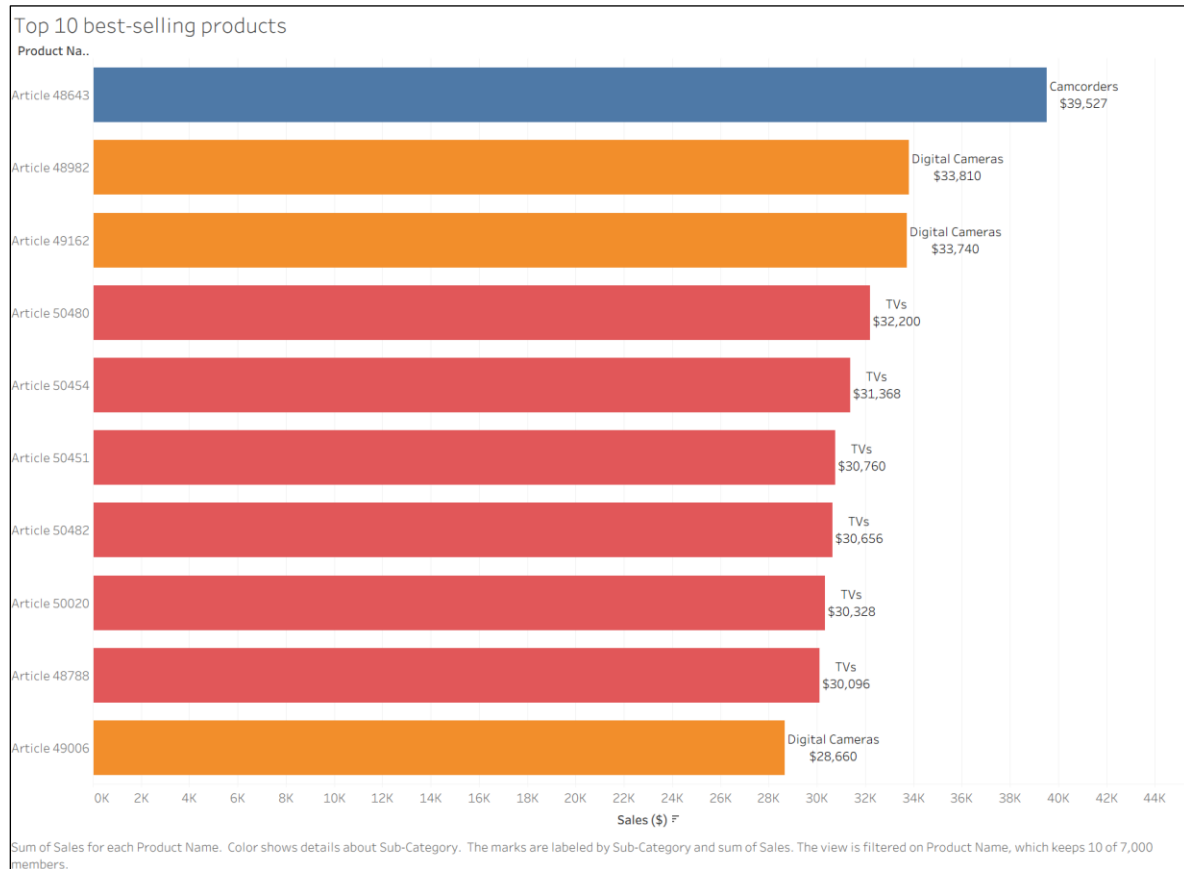
3.2 Sales by product sub-category

From this bar chart, TV is the highest seller (\$4,511,224) in the TV category and among all the other sub-categories as well. The leading sub-category for Camera is Digital Cameras (\$3,644,900). The leading sub-category for Audio & HiFi is Speakers (\$4,469,708). Across all categories, their respective accessories have the lowest sale figures. This suggests that consumers often bought the main products first and accessories are bought later or less frequently.



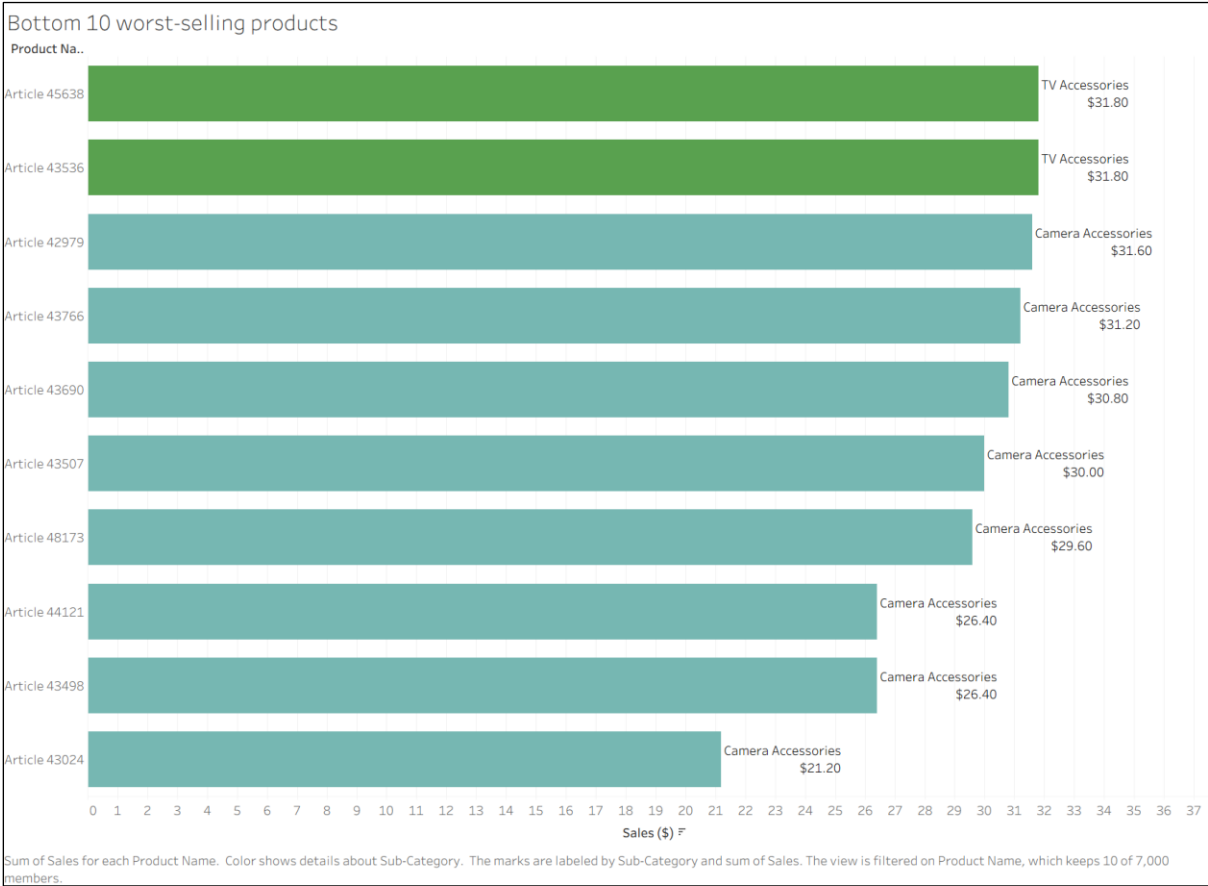
3.3 Top 10 best-selling products

Overall, Article 48643 (Camcorder) is the top-performing product with sales of \$39,427, followed by Digital Cameras and TVs. These are high-value tech products that are often purchased without their accessories.



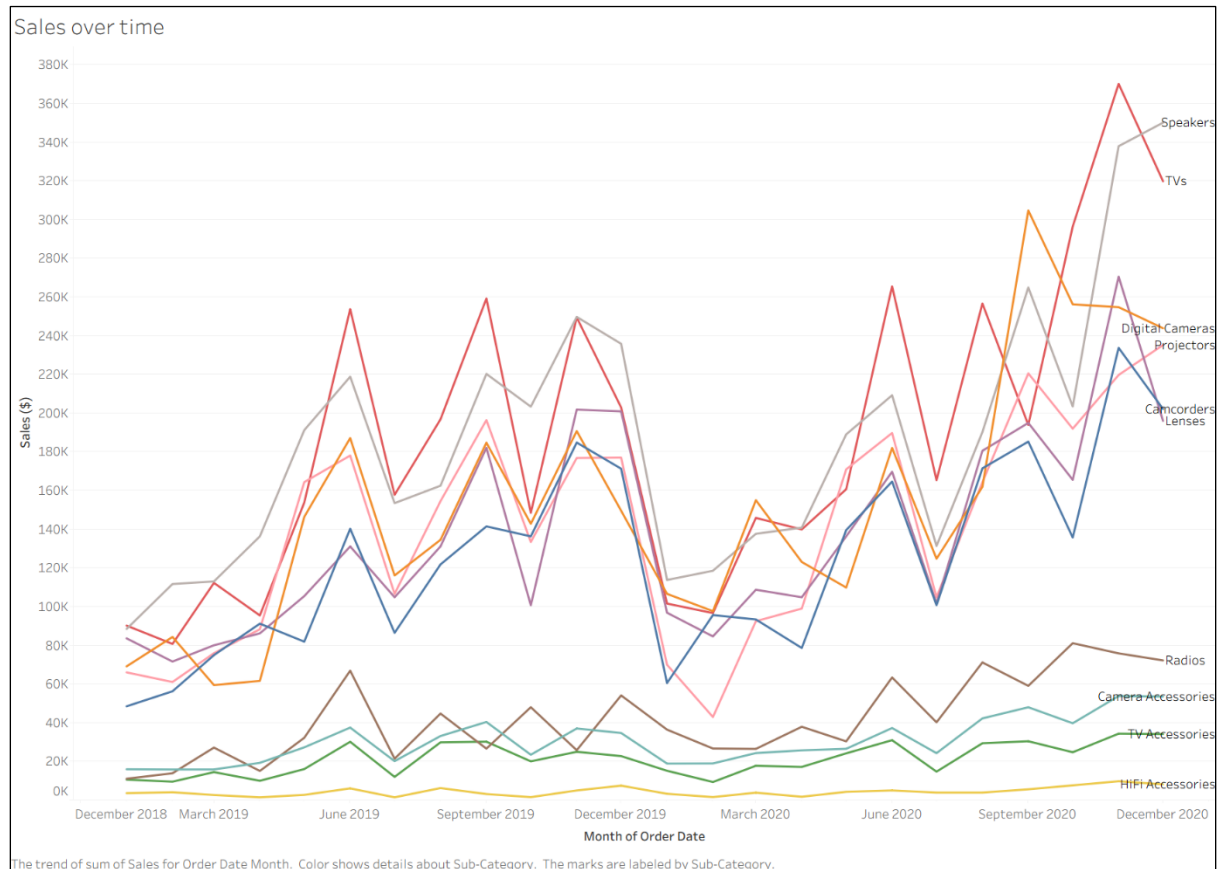
3.4 Bottom 10 worst-selling products

Article 43024 (Camera Accessories) dominates the bottom of the sales chart, suggesting that this product is performing poorly. There is a narrow sales range of \$21 to \$32 for all the products listed here, indicating that they are performing poorly at a similar level. These are smaller add-ons products that may be neglected.



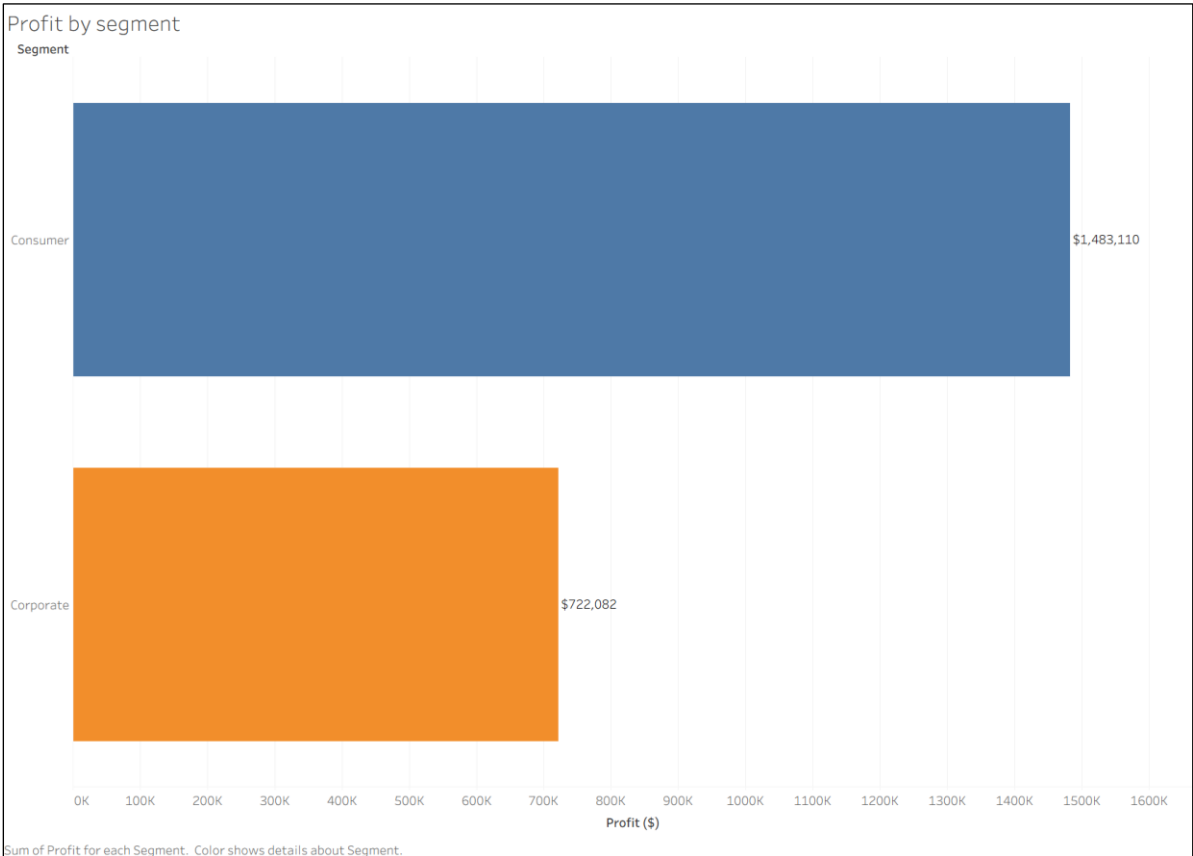
3.5 Sales over time by product sub-categories

Speakers and TVs dominate in total sales and growth rate while Digital Camera and Camcorders show increasing trend and demand. There are also visible seasonal spikes in late 2019 and from mid to late 2020. However, Accessories (Camera, TV, HiFi) remain generally flat despite the growth in main products.



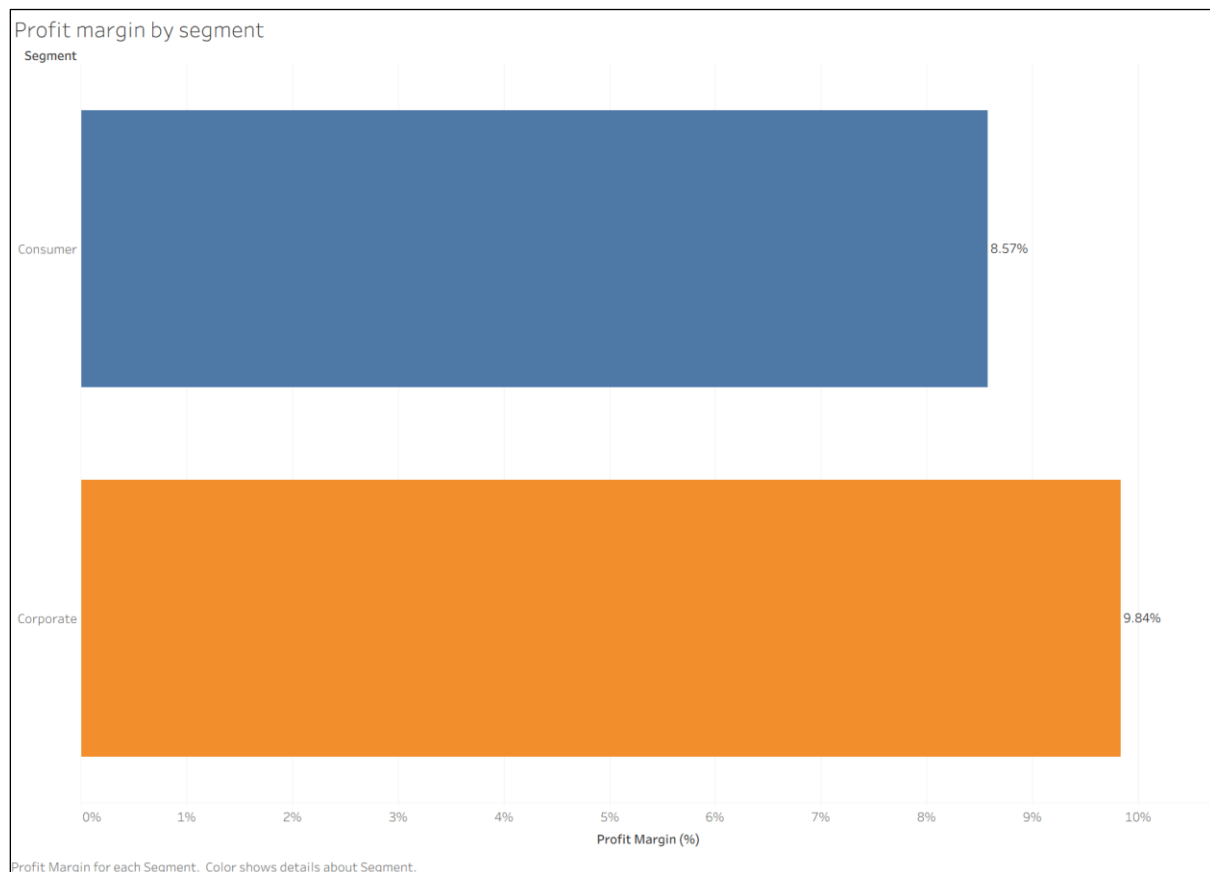
3.6 Profit by segment

The bar chart below shows the total profit generated by each customer's segment. Consumer segment generated a higher profit of \$1,483,110 compared to the corporate segment, which generated \$722,082.



3.7 Profit margin by segment

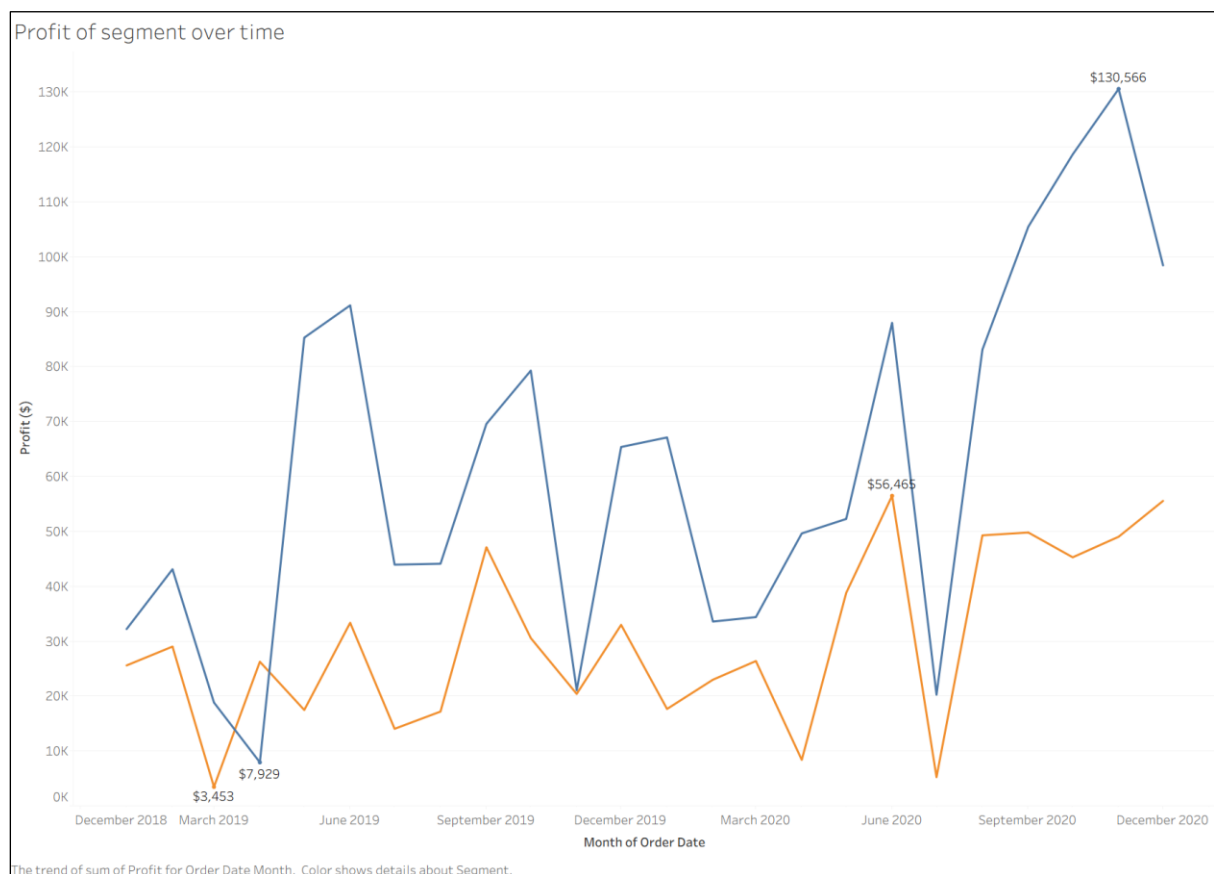
However, the corporate segment has a slightly higher profit margin (9.84%) as compared to the consumer segment (8.57%). This suggests that corporate is more profitable per dollar sold and that the consumer segment might be less efficient in profitability, but it still makes more profit because of its higher overall sales volume.



3.8 Profit of segment over time

The consumer segment has a higher overall profit with sharp spikes and drops. It peaks in November 2020 (\$130,566) and other high points were seen around mid-2019 and mid-2020. The biggest drop was in April 2019 (\$7,929). The seasonal sales and promotions might have affected the buying behavior of the consumer and therefore, they are less predictable and can vary a lot.

It makes sense that the profit for corporate segment is lower but much more stable. It peaks in June 2020 (\$56,465) and the lowest point would be in March 2019 (\$3,453). This is probably because corporate buyers usually purchase products based on contracts or agreements. Their behavior is predictable and unlikely to be affected by sudden changes or trends which explains why the line is steadier.



3.9 Profit for each product sub-category by segment

The highest profit generated from consumer would be TVs (\$304,463), Digital Cameras (\$279,231), Speakers (\$275,683), Projectors (\$245,933) and Camcorders (\$211,631). The profit is negative only for TV Accessories (-\$9,348). Consumer tends to generate higher profits across most sub-categories compared to corporate customers.

The highest profit generated from corporate customers would be Digital Cameras (\$182,483), TVs (\$161,905), and Projectors (\$119,656). The profit is negative only for TV Accessories (-\$4,960).

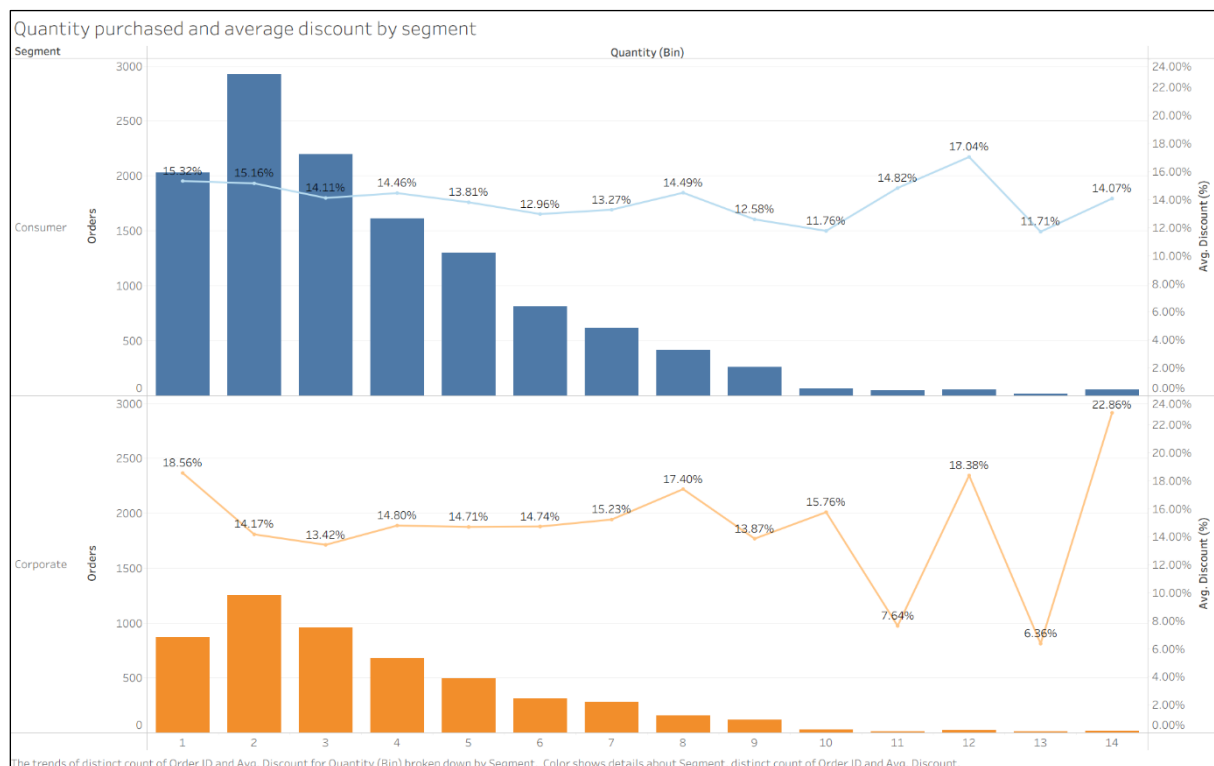
Overall, Digital Cameras and TVs generate the highest profits for both consumer as well as corporate customers.



3.10 Quantity purchased and average discount

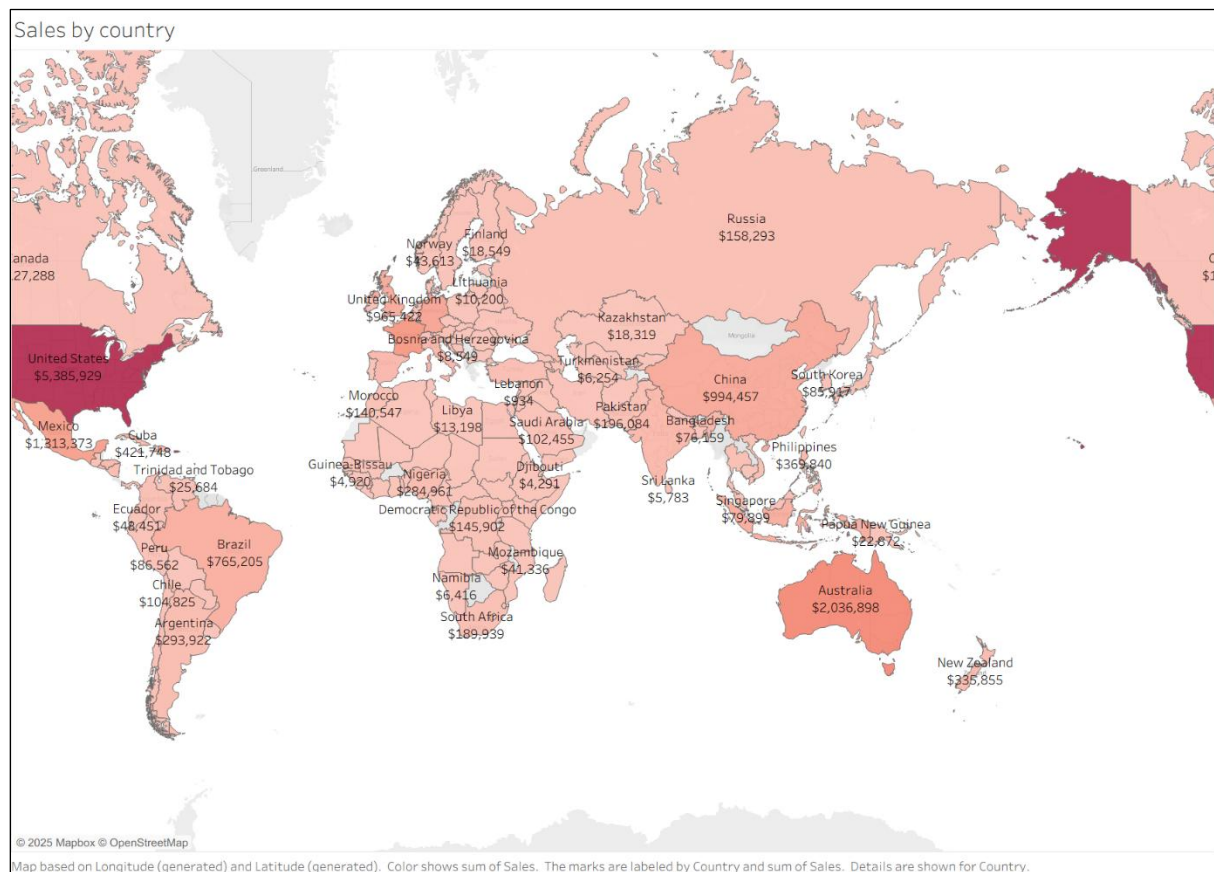
This is a distribution of quantity purchased by customer segment. Most orders were purchased in quantities of 1 to 4 units. Both the consumer and corporate segment follow a similar right-skewed distribution shape. However, consumer orders generally tend to have higher order quantities than corporate orders. Higher quantities of 6 and above drop off quickly, suggesting that bulk buying is uncommon, especially for corporate customers.

Consumer has an average discount that typically hovers around 13% to 16% with its highest point at quantity 12 (17.04%). However, the average discount for corporate customers fluctuates more from 6.38% at quantity 13 to 22.86% at quantity 14. This suggests that the highest average discounts in both segments occur at larger quantities, most probably to encourage bulk buying or clear inventory. From this information, the management can optimize their discount strategies by knowing that most customers buy in small quantities. For example, volume discounts for corporates customers and bundle deals for consumers would increase the quantities purchased.



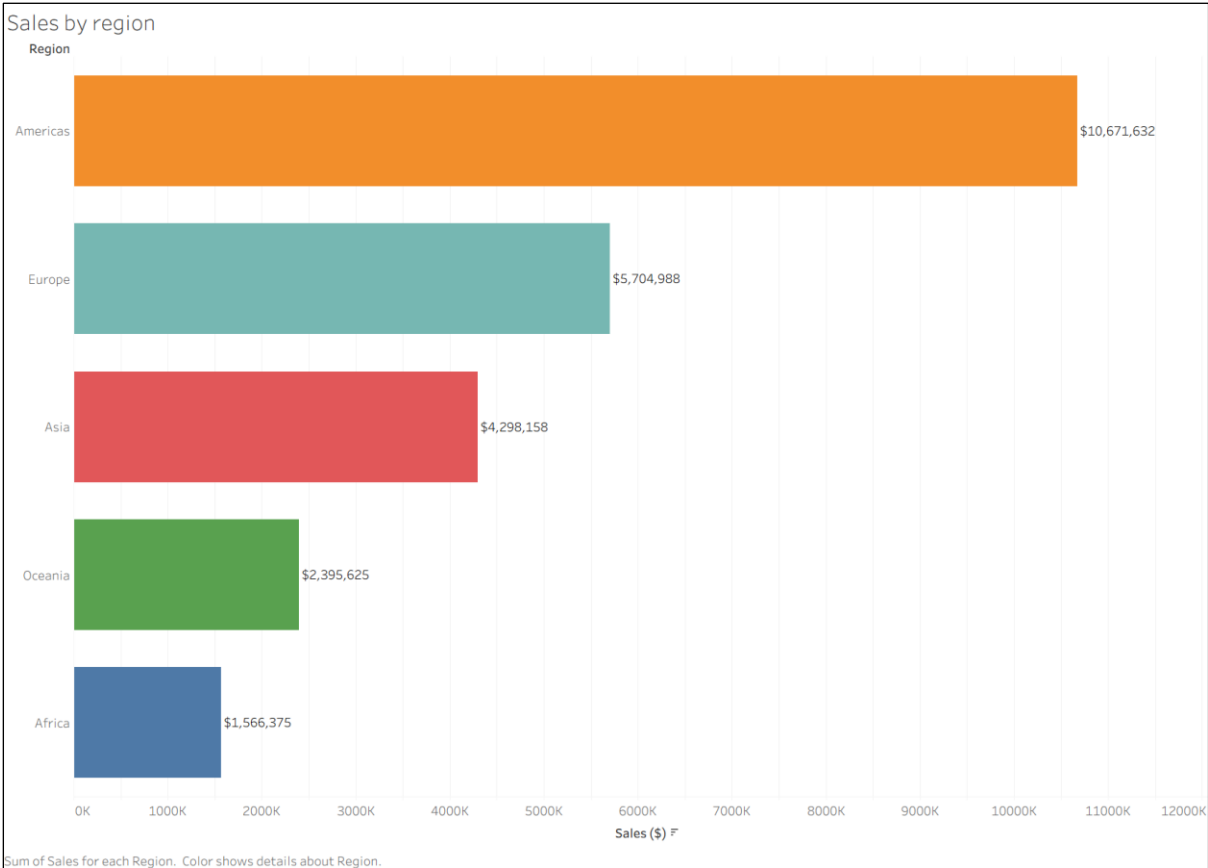
3.11 Sales across countries

The highest sales were generated in the United States (\$5,385,929). There were also significant sales in Australia (\$2,036,898), France (\$1,521,658), Mexico (\$1,313,373) and Germany (\$1,148,722). Lower sales are in many African countries and Southeast Asian countries.



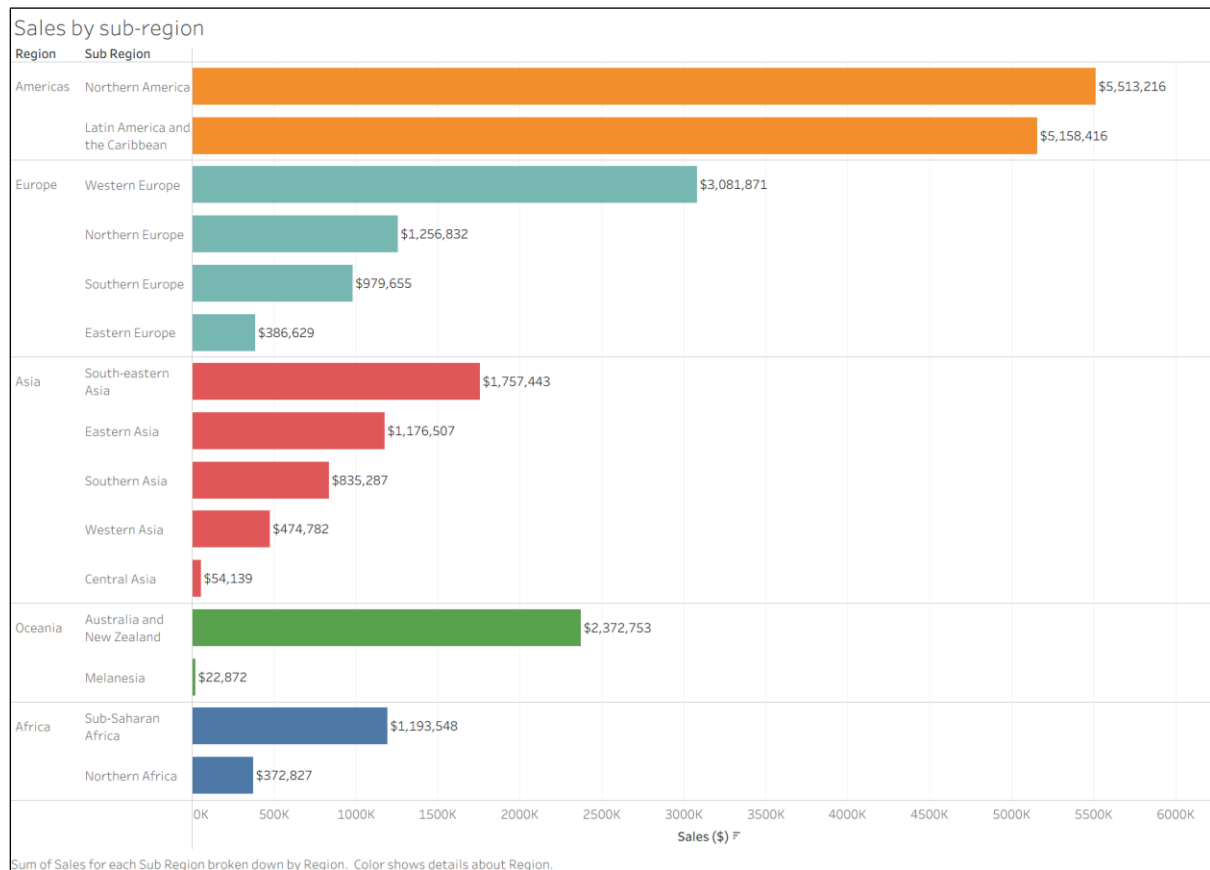
3.12 Sales across regions

Americas (\$10,671,632) is the dominant sales region followed by Europe and Asia. Although Oceania and Africa are trailing behind, it represents potential growth.



3.13 Sales across sub-regions

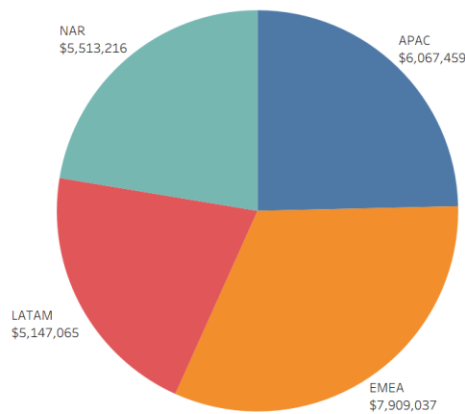
Diving deeper into sub-regions, Northern America contributed the most sales (\$5,513,216), while Melanesia recorded the lowest sales at \$22,872. There are several regions that are showing high-growth potential, and the management can consider planning for an expansion. As for those regions that are underperforming, evaluate whether to scale back and shift efforts to stronger regions.



3.14 Sales across market

This pie chart shows the sales distribution across four markets. EMEA leads with \$7.9 million in sales, followed by APAC with \$6.1 million, NAR with \$5.5 million and LATAM being the lowest performer among the four. The sales distribution is also quite balanced, showing strong contributions from all regions.

Sales by market

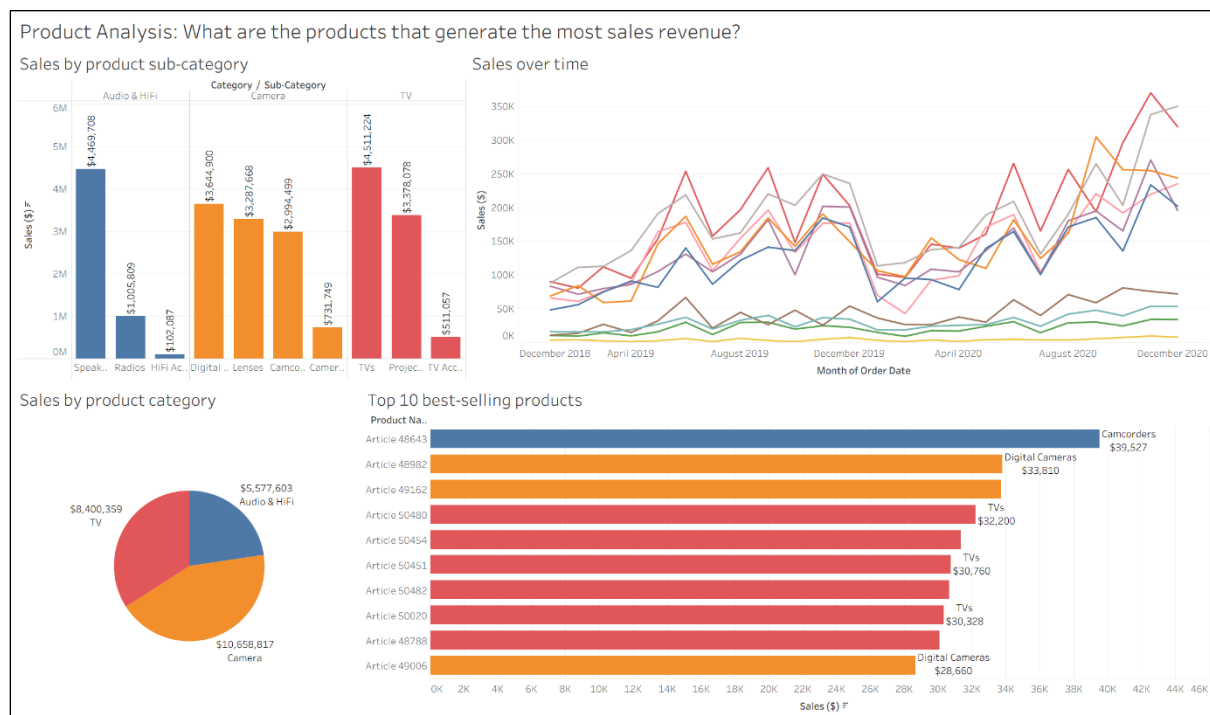


Market and sum of Sales. Color shows details about Market. Size shows sum of Sales. The marks are labeled by Market and sum of Sales.

4. Dashboard

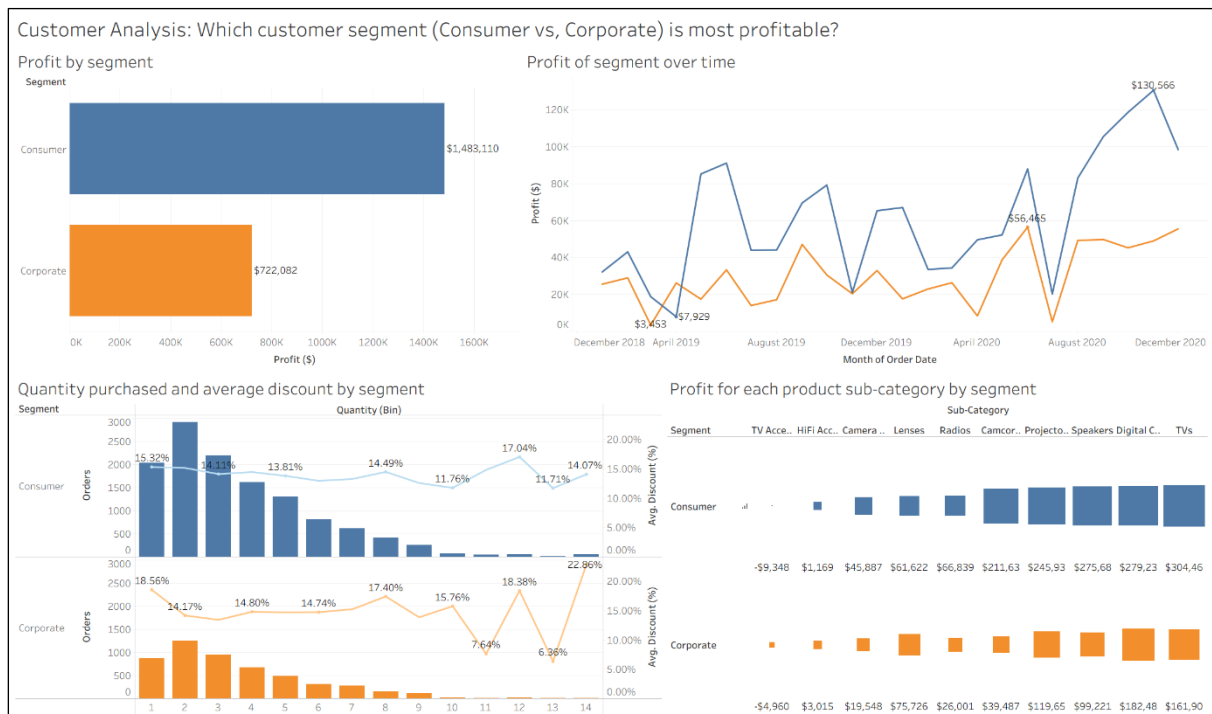
4.1 Product Analysis: What are the products that generate the most sales revenue?

Article 48643 (Camcorder) is the top-selling product overall with sales of \$39,527. Article 48982 & 49162 (Digital Camera) are strong performers while the rest are TVs. The management should focus efforts on the best sellers in making sure there are consistent stock and smooth supply chain for these top products. Despite the high sales of main products, accessories are underperforming. One possible way to increase the sales of accessories is to provide bundle discounts for buying the main product along with its accessories. The management should also identify non-selling accessories for possible discontinuation or consider replacing them with in-demand products.



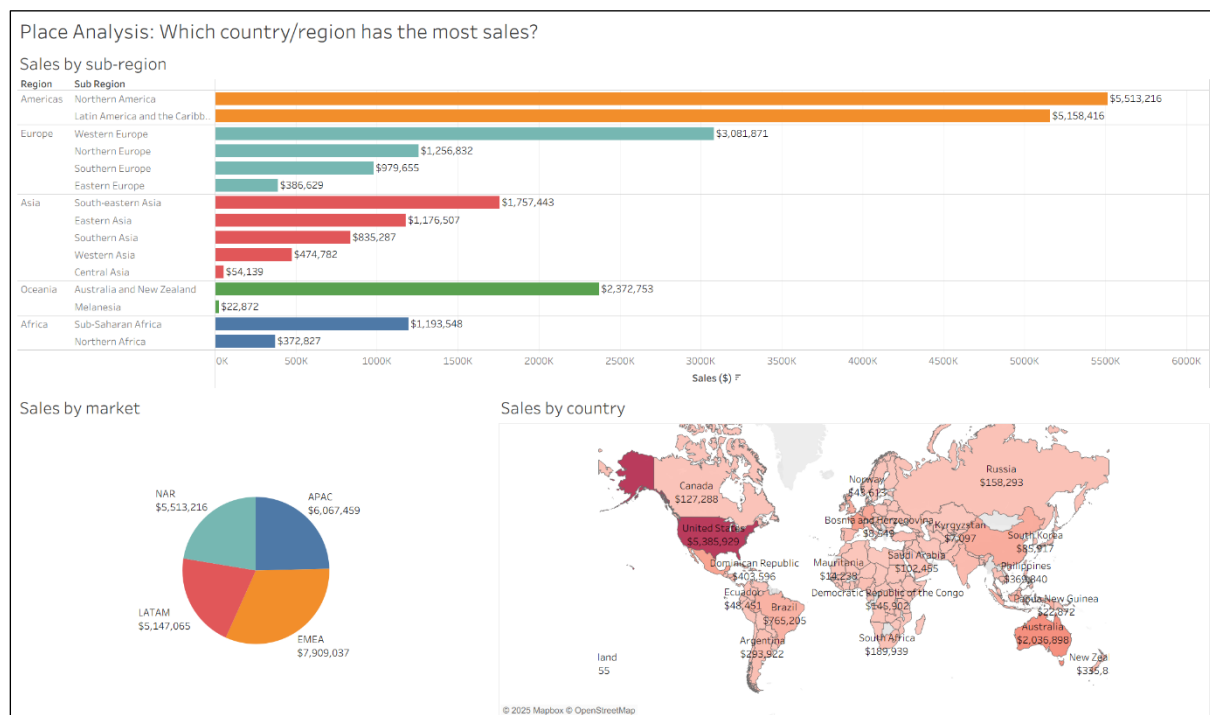
4.2 Customer Analysis: Which customer segment (Consumer vs. Corporate) is most profitable?

Overall, the consumer segment is more profitable than the corporate segment. However, the corporate segment is only underperforming slightly. The management might want to close the gap with adjusted strategies to encourage larger, more frequent purchases from corporate customers. They can either focus on one segment only or take this opportunity to grow both segments and achieve more by taking a balanced approach.



4.3 Place Analysis: Which country/region has the most sales?

United States and the broader Americas region lead in sales. The management should strengthen the core market and protect them as they are important for sustaining revenue. Despite the Americas being the top-performing region, market such as EMEA and APAC also perform strongly. The management can see this as a potential growth to diversify their global presence and avoid relying on a single region only. Other than that, regions like Melanesia and Central Asia should be evaluated whether the management should continue investing in them.



References

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